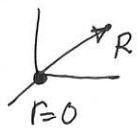
Electrostatics of point charges and conductors

$$\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\varepsilon_0}, \quad \vec{\nabla}\times\vec{E} = 0 \quad \text{(static electric field)}$$

$$\vec{\nabla}\cdot\vec{B} = 0, \quad \vec{\nabla}\times\vec{B} = \mu_0\vec{j}$$

Superposition principle (consequence of linearity of Maxwell's eqns)

$\Rightarrow$ start with $\rho(r) = q\,\delta(\vec{r})$



Only special vector $\vec{R}$ from q to the point of observation $\Rightarrow \vec{E}(\vec{R}) = \vec{R}\,f(R)$

$R = |\vec{R}|$

Gauss's law $\displaystyle\int_S d\vec{A}\cdot\vec{E} = \frac{q}{\varepsilon_0}$

$$4\pi R^2\,\cancel{d\vec{R}}\cdot R\,f(R) = \frac{q}{\varepsilon_0} \Rightarrow f = \frac{q}{4\pi\varepsilon_0 R^3}$$

$$\vec{E}(\vec{R}) = \frac{q\,\vec{R}}{4\pi\varepsilon_0 R^3}$$

$$\vec{E}(\vec{R}) = -\vec{\nabla}\varphi \Rightarrow \varphi(\vec{R}) = \frac{q}{4\pi\varepsilon_0 R}$$

since $\displaystyle\vec{\nabla}\frac{1}{R} = \frac{\partial}{\partial\vec{R}}\frac{1}{R} = -\frac{1}{R^2}\frac{\partial R}{\partial\vec{R}} = -\frac{\vec{R}}{R^3}$.

$$\frac{\partial R}{\partial R_i} = \frac{R_i}{R} \quad \Longrightarrow \quad \frac{\partial R}{\partial\vec{R}} = \frac{\vec{R}}{R}$$

Many particles

$$\vec{E}(\vec{R}) = \frac{1}{4\pi\varepsilon_0}\sum_n \frac{q_n(\vec{R}-\vec{r}_n)}{|\vec{R}-\vec{r}_n|^3} \Rightarrow \varphi(\vec{R}) = \frac{1}{4\pi\varepsilon_0}\sum_n \frac{q_n}{|\vec{R}-\vec{r}_n|}$$

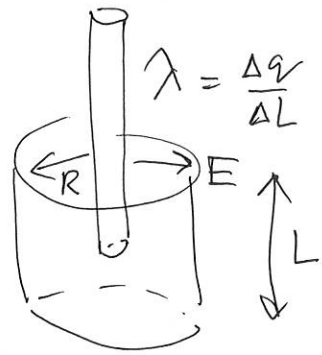
n ← particle index

$\downarrow$ continuum

$$\vec{E}(\vec{R}) = \frac{1}{4\pi\varepsilon_0}\int dv\,\frac{\rho(\vec{r})(\vec{R}-\vec{r})}{|\vec{R}-\vec{r}|^3}, \quad \varphi(\vec{R}) = \frac{1}{4\pi\varepsilon_0}\int dv\,\frac{\rho(\vec{r})}{|\vec{R}-\vec{r}|}$$

Basic Gauss law

Linear charge $\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\varepsilon_0} \Rightarrow \oint d\vec{a}\cdot\vec{E} = \frac{1}{\varepsilon_0}\int dV \rho$



$$\lambda = \frac{\Delta q}{\Delta L}$$

$$E \cdot 2\pi R L = \frac{\lambda L}{\varepsilon_0}$$

$$\Rightarrow E = \frac{\lambda}{2\pi\varepsilon_0 R}$$

$$\Rightarrow \phi(R) = \frac{\lambda}{2\pi\varepsilon_0}\ln\left(\frac{R_0}{R}\right)$$

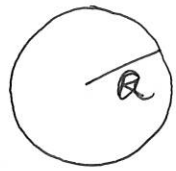
In cylindrical coordinates $\vec{\nabla} t = \frac{\partial t}{\partial \rho}\hat{\rho} + \frac{1}{\rho}\frac{\partial t}{\partial \phi}\hat{\phi} + \frac{\partial t}{\partial z}\hat{z}$.

and $\rho = R$ here.

Spherical: $\vec{\nabla} t = \frac{\partial t}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial t}{\partial \theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial t}{\partial \phi}\hat{\phi}$

Thin charged shell : uniformly charged shell

surface density $\sigma = \frac{Q}{4\pi a^2}$.



$$\Rightarrow \rho(r) = \sigma\, \delta(r-a)$$

Check: $Q = \int dV \rho(r) = 4\pi\sigma \int dr\, r^2 \cdot \delta(r-a) = 4\pi a^2 \sigma.$

Laplacian: $$\nabla^2 t = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial t}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial t}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 t}{\partial \varphi^2}$$

By spherical symmetry here $\phi(\vec{r}) = \phi(r)$

$$\Rightarrow \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \phi}{\partial r}\right) = -\frac{\sigma}{\varepsilon_0}\delta(r-a)$$

Physics: $\vec{E}$ inside $= 0$, $\vec{E}$ outside $\sim \frac{\hat{r}}{r^2}$

$\Downarrow$ $\phi(r) = \text{const}$, $r<a$. $\qquad$ $\Downarrow$ $\phi(r) \sim \frac{1}{r}$, $r>a$.

$$\int_{a-\eta}^{a+\eta} dr\, \frac{d}{dr}\left(r^2 \frac{d\phi}{dr}\right) = -\frac{\sigma}{\varepsilon_0}\int_{a-\eta}^{a+\eta} dr\, r^2 \delta(r-a) = -\frac{\sigma a^2}{\varepsilon_0}$$

Note: it is η-independent

$$\lim_{\eta \to +0}\left(r^2 \frac{d\phi}{dr}\Big|_{a-\eta}^{a+\eta}\right) = a^2\left(\frac{d\phi}{dr}\Big|_{a+0} - \frac{d\phi}{dr}\Big|_{a-0}\right) = -\frac{\sigma a^2}{\varepsilon_0}$$

$$\Rightarrow \phi'(a+0) - \phi'(a-0) = -\frac{\sigma}{\varepsilon_0}$$

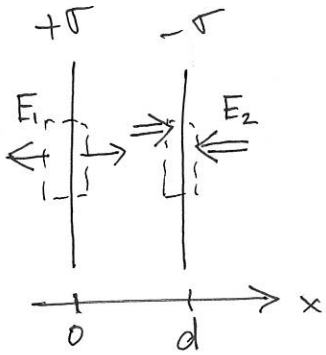
$$\Rightarrow E(a+0) - E(a-0) = \frac{\sigma}{\varepsilon_0}$$

Electric field is discontinous, $E(a+0) - E(a-0) = \frac{\sigma}{\varepsilon_0}$.
but electric potential is continuous, $\varphi(a+0) = \varphi(a-0)$.

Why: discontinuous $\phi(r)$ implies infinite electric field.
This is not possible/physical but something like that can happen at the <u>double layer</u>

$2E_1 A = \frac{\sigma A}{\varepsilon_0}$

$E_1 = \frac{\sigma}{2\varepsilon_0}$

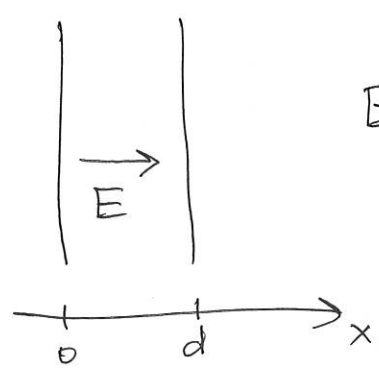


$E_2 = -\frac{\sigma}{2\varepsilon_0}$

* so that $E_1(+0) - E_1(-0) = \frac{\sigma}{2\varepsilon_0} - \left(-\frac{\sigma}{2\varepsilon_0}\right) = \frac{\sigma}{\varepsilon_0}$ just as we found for the thin shell.

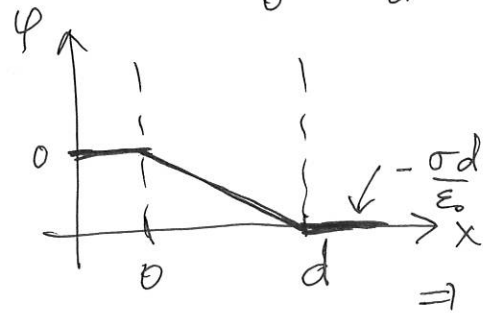
(The difference to that for the shell $E(a-0) = 0$, all field is outside.)

The net result



$E = 2 \cdot \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$.

$\Rightarrow$



$\varphi(x) = -Ex = -\frac{\sigma}{\varepsilon_0} x$

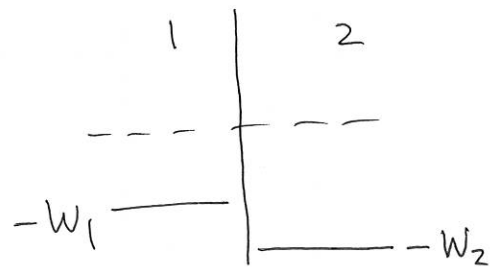
for $x \in (0, d)$

$\Rightarrow \varphi(0) - \varphi(d) = \frac{\sigma d}{\varepsilon_0}$ = finite potential jump

Physical realization of double layer potential:
contact between two materials with different work function W_1 and W_2.

Work function W = energy required to remove electron from the material. $W > 0$ and large.

Suppose $W_1 < W_2$



$\Rightarrow$ electrons from ① would like to flow to ②

$\Rightarrow$ positive charge builds on ①, negative − on ②.

The flow stops when $(-e)\Delta\phi = W_1 - W_2$.

$$\frac{e\sigma d}{\varepsilon_0} = W_2 - W_1 .$$

Typically $d \sim$ atomic distances $\Rightarrow$ ~~potential~~ electric field inside the double layer is large (but finite).

Space-charge limited current (P3.1)

$$\nabla^2 \phi = -\frac{\rho}{\varepsilon_0} \qquad v\rho = J = \text{const} = -e\, v(x)\, n(x) = -J_A$$

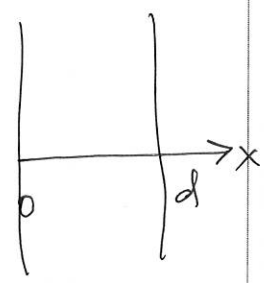
$$E = \frac{1}{2} m v_0^2 = \frac{1}{2} m v^2(x) - e\phi(x) = 0$$

$$v(x) = \sqrt{\frac{2e\phi(x)}{m}}$$

$$\frac{d^2}{dx^2}\phi = +\frac{J_A}{v(x)\varepsilon_0} = +\sqrt{\frac{m}{2e}}\,\frac{J_A}{\varepsilon_0}\,\frac{1}{\phi^{1/2}}$$

$$\cancel{J = -e\,n(x)\,v(x) = -J}$$

$$\underbrace{\frac{d\phi}{dx}\,\frac{d^2\phi}{dx^2}} = +C\,\frac{1}{\phi^{1/2}}\,\frac{d\phi}{dx}$$



$$\frac{d}{dx}\left[\frac{1}{2}\left(\frac{d\phi}{dx}\right)^2\right] = +2C\,\frac{d\phi^{1/2}}{dx}$$

$$\Rightarrow \frac{1}{2}\left(\frac{d\phi}{dx}\right)^2 = +2C\,\phi^{1/2} + C_0$$

BC: $\phi(x=0) = 0$, $\vec{E}(x=0) = 0$

$$\Rightarrow C_0 = 0$$

$$\frac{d\phi}{dx} = \sqrt{4C}\,\phi^{1/4}$$

$$\int_0^x \frac{d\phi}{\phi^{1/4}} = \sqrt{4C}\int_0^x dx$$

$$\frac{4}{3}\int d\phi^{3/4} = \sqrt{4C}\,x$$

$$\Rightarrow \frac{4}{3}\phi^{3/4}(x) = 2\sqrt{C}\,x$$

$$\phi^{3/4}(x) = \frac{3}{2}\sqrt{C}\,x = \frac{3}{2}\left(\frac{m}{2e}\right)^{1/4}\left(\frac{J_A}{\varepsilon_0}\right)^{1/2} x$$

$$\phi(x) = \left(\frac{3}{2}\right)^{4/3}\left(\frac{m}{2e}\right)^{\frac{1}{3}}\left(\frac{J_A}{\varepsilon_0}\right)^{2/3} x^{4/3}$$

$$\Rightarrow \quad \frac{-(-e)\,n(x)}{\varepsilon_0} = \nabla^2\phi = \frac{d^2\phi}{dx^2} \qquad \phi = A\,x^{4/3}$$

$$e\,n(x) = \varepsilon_0 A \frac{d^2\, x^{4/3}}{dx^2} = \frac{4}{3}\varepsilon_0 A \frac{d\,x^{1/3}}{dx} = \frac{4}{9}\varepsilon_0 A \frac{1}{x^{2/3}}$$

$$x^{2/3} = \left(\frac{\phi}{A}\right)^{1/2} \Rightarrow e\,n(x) = \frac{4}{9}\varepsilon_0 A \left(\frac{A}{\phi}\right)^{1/2}$$

$$\Rightarrow J = e\,n(x)\,v(x) = \frac{4}{9}\varepsilon_0 A \left(\frac{A}{\phi}\right)^{1/2} \cdot \sqrt{\frac{2e}{m}\phi} = \text{Const.}$$

At $x = d$ $\phi(d) = V_A = B\,d^{4/3}\,J^{2/3} \Rightarrow J \sim V_A^{3/2}$.